

Ordinals and Typed Lambda Calculus

Ordinal Notations

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2018-2

Introduction

Too many numbers to be named

Spector (1955) starts by pointing out that

*Cantor's second ordinal number class is perhaps the simplest example of a set of mathematical objects which cannot all be named in **one** language.*

Introduction

Remark

Recall that a language is a subset of words over an alphabet.

Introduction

Remark

Recall that a language is a subset of words over an alphabet.

Question

Can all natural numbers be named in one language?

Introduction

Remark

Recall that a language is a subset of words over an alphabet.

Question

Can all natural numbers be named in one language?

Yes! We can name any natural number by a word over the alphabet $\{0, 1, 2, \dots, 9\}$.

Introduction

Remark

Recall that a language is a subset of words over an alphabet.

Question

Can all natural numbers be named in one language?

Yes! We can name any natural number by a word over the alphabet $\{0, 1, 2, \dots, 9\}$.

That was easy because the set of natural numbers is denumerable.

Introduction

Question

Can all real numbers be named in one language?

Notation Systems for the Ordinals below ϵ_0

Theorem

Every ordinal α less than ϵ_0 has a normal form

$$\alpha = \omega^{\beta_1} + \omega^{\beta_2} + \cdots + \omega^{\beta_n},$$

where $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$ are ordinals and $\beta_i < \alpha$ (Pohlers 2009, p. 33).

Notation Systems for the Ordinals below ϵ_0

Theorem

Every ordinal α less than ϵ_0 has a normal form

$$\alpha = \omega^{\beta_1} + \omega^{\beta_2} + \cdots + \omega^{\beta_n},$$

where $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$ are ordinals and $\beta_i < \alpha$ (Pohlers 2009, p. 33).

Definition

From the above theorem, we can name any ordinal less than ϵ_0 by a word over the alphabet $\{+, 0, \omega\}$. By coding these words in natural numbers, we get a **notation system for the ordinals below ϵ_0** (Pohlers 2009, p. 33).

Notation Systems for the Ordinals below ϵ_0

Theorem

Every ordinal α less than ϵ_0 has a normal form

$$\alpha = \omega^{\beta_1} + \omega^{\beta_2} + \cdots + \omega^{\beta_n},$$

where $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$ are ordinals and $\beta_i < \alpha$ (Pohlers 2009, p. 33).

Definition

From the above theorem, we can name any ordinal less than ϵ_0 by a word over the alphabet $\{+, 0, \omega\}$. By coding these words in natural numbers, we get a **notation system for the ordinals below ϵ_0** (Pohlers 2009, p. 33).

Remark

The ordinal ϵ_0 is the smallest ordinal that has no a name in terms of ω .

Notation Systems for the Ordinals below ϵ_0

Representation using trees

We can also represent ordinals below ϵ_0 using finite (ordered) trees.

Notation Systems for the Ordinals below ϵ_0

Representation using trees

We can also represent ordinals below ϵ_0 using finite (ordered) trees.

Theorem

There is a one-to-one correspondence between finite rooted trees and ordinals below ϵ_0 given by (Dershowitz 1993):

- i) The one-node tree represents the ordinal 0.
- ii) The tree with sub-trees representing the ordinals $\alpha_1, \dots, \alpha_n$ represents the ordinal $\omega^{\alpha_1} \# \dots \# \omega^{\alpha_n}$.

Kleene's $\mathcal{O}$

Definition

We inductively define the **notation system** $\mathcal{O}$ and the **well-founded ordering** $<_{\mathcal{O}}$ (Cooper 2004, Definition 16.2.29, p. 358):

- i) We start by giving the ordinal 0 notation 1. Assume all ordinals less than α have been assigned notations, and $<_{\mathcal{O}}$ has been defined on these notations.
- ii) Say $\alpha = \beta + 1$, and β has notation x .

Then α gets notation 2^x and we add $\langle z, 2^x \rangle$ to $<_{\mathcal{O}}$ for each z such that $z = x$ or $z <_{\mathcal{O}} x$ already.

- iii) Say α is a limit ordinal, and $\langle \varphi_e(n) : n \in \mathbb{N} \rangle$ is a list of notations for ordinals with limit α , and $\forall n [\varphi_e(n) <_{\mathcal{O}} \varphi_e(n+1)]$ already.

Then give α notation $3 \cdot 5^e$, and add $\langle z, 3 \cdot 5^e \rangle$ to $<_{\mathcal{O}}$ for all z for which $z <_{\mathcal{O}} \varphi_e(n)$ already, some $n \geq 0$.

Kleene's $\mathcal{O}$

Remark

We could use the notations zero , $\text{succ}(x)$ and $\text{lim}(e)$ instead of the notations 1 , 2^x and $3 \cdot 5^e$.[†]

[†]See, e.g. (Fránzen [2004] 2017).

Constructive Ordinals and Computable Ordinals

Definition

The **constructive ordinals** (second definition) are the ordinals notated by $\mathcal{O}$ (Cooper 2004, Definition 16.2.29, p. 358).

[†]See, e.g. (Cooper 2004, Definition 16.2.25, p. 358), (Rogers [1967] 1992, p. 211) and (Ash and Knight 2000, p. 61).

Constructive Ordinals and Computable Ordinals

Definition

The **constructive ordinals** (second definition) are the ordinals notated by $\mathcal{O}$ (Cooper 2004, Definition 16.2.29, p. 358).

Definition

A countable ordinal is **computable** iff it is finite or it is isomorphic to a computable well-ordering $(A, \prec)$.[†]

[†]See, e.g. (Cooper 2004, Definition 16.2.25, p. 358), (Rogers [1967] 1992, p. 211) and (Ash and Knight 2000, p. 61).

Constructive Ordinals and Computable Ordinals

Theorem

An ordinal α is constructive iff α is a computable ordinal.[†]

[†]See, e.g. (Rogers [1967] 1992, Corollary XIX and Theorem XX, p. 211) and (Ash and Knight 2000, § 4.7, p. 62).

References

- C. J. Ash and J. Knight (2000). Computable Structures and the Hyperarithmetical Hierarchy. Vol. 144. Studies in Logic and the Foundations of Mathematics. Elsevier (cit. on pp. 15–17).
- S. Barry Cooper (2004). Computability Theory. Chapman & Hall (cit. on pp. 13, 15, 16).
- Nachum Dershowitz (1993). Trees, Ordinals and Termination. In: Theory and Practice of Software Development (TAPSOFT'93). Ed. by Marie-Claude Gaudel and Jean-Pierre Jouannaud. Vol. 668. Lecture Notes in Computer Science. Springer, pp. 243–250. DOI: [10.1007/3-540-56610-4_68](https://doi.org/10.1007/3-540-56610-4_68) (cit. on pp. 11, 12).
- Torkel Fránzen [2004] (2017). Inexhaustibility. A Non-Exhaustive Treatment. Vol. 16. Lecture Notes in Logic. Association for Symbolic Logic and Cambridge University Press. DOI: [10.1017/9781316755969](https://doi.org/10.1017/9781316755969) (cit. on p. 14).
- Wolfram Pohlers (2009). Proof Theory. The First Step into Impredicativity. Springer. DOI: [10.1007/978-3-540-69319-2](https://doi.org/10.1007/978-3-540-69319-2) (cit. on pp. 8–10).
- Hartley Rogers [1967] (1992). Theory of Recursive Functions and Effective Computability. Third printing. MIT Press (cit. on pp. 15–17).
- Clifford Spector (1955). Recursive Well-Orderings. The Journal of Symbolic Logic 20.2, pp. 151–163. DOI: [10.2307/2266902](https://doi.org/10.2307/2266902) (cit. on p. 2).